ACS Variable Reference Guide

Date: 2026-01-08
Purpose: Reference guide for ACS (American Community Survey) variables and metrics used in tenant cohort stratification

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Overview

This document provides a comprehensive reference for ACS variables and metrics used in STRATA's tenant cohort stratification strategy. ACS data is sourced from the U.S. Census Bureau's American Community Survey and provides demographic, economic, and housing characteristics at the ZIP and CBSA levels.

Part 1: Income Variables

B19013: Median Household Income

Variable Code: `B19013_001`
Description: Median household income in the past 12 months (inflation-adjusted dollars)
Geography Levels: ZIP, CBSA
Use Case: Primary income stratification metric
Stratification:

- `HIGH_INCOME`: >= 100K
- `MIDDLE_INCOME`: 50K-100K
- `LOW_INCOME`: < 50K

B19001: Household Income Distribution

Variable Codes:

- `B19001_002`: <10K
- `B19001_003`: 10K-14,999
- `B19001_004`: 15K-19,999
- `B19001_005`: 20K-24,999
- `B19001_006`: 25K-29,999
- `B19001_007`: 30K-34,999
- `B19001_008`: 35K-39,999
- `B19001_009`: 40K-44,999
- `B19001_010`: 45K-49,999
- `B19001_011`: 50K-59,999
- `B19001_012`: 60K-74,999
- `B19001_013`: 75K-99,999
- `B19001_014`: 100K-124,999
- `B19001_015`: 125K-149,999
- `B19001_016`: 150K-199,999

- `B19001_017`: 200K+

Description: Distribution of households by income brackets
Geography Levels: ZIP, CBSA
Use Case: Income distribution analysis, income cohort stratification
Derived Metrics:

- `PCT_INCOME_<BRACKET>`: Percentage of households in each income bracket
- `PCT_MIDDLE_INCOME`: Sum of 50K-150K brackets
- `PCT_HIGH_INCOME`: Sum of 100K+ brackets

Part 2: Household Type Variables

B11001: Household Type

Variable Codes:

- `B11001_002`: Family households
- `B11001_007`: Non-family households

Description: Distribution of family vs non-family households
Geography Levels: ZIP, CBSA
Use Case: Household type stratification
Derived Metrics:

- `PCT_FAMILY_HOUSEHOLDS`: Percentage of family households
- `PCT_NON_FAMILY_HOUSEHOLDS`: Percentage of non-family households

B11016: Household Type by Presence of Children

Variable Codes:

- `B11016_002`: Family households with children
- `B11016_010`: Family households without children
- `B11016_011`: Non-family households

Description: Household type broken down by presence of children
Geography Levels: ZIP, CBSA
Use Case: Family-oriented tenant cohort identification

Household Size Distribution

Derived from B11001 and related tables:

- `PCT_HOUSEHOLD_SIZE_1`: Single-person households
- `PCT_HOUSEHOLD_SIZE_2`: Two-person households
- `PCT_HOUSEHOLD_SIZE_3`: Three-person households
- `PCT_HOUSEHOLD_SIZE_4`: Four-person households
- `PCT_HOUSEHOLD_SIZE_5PLUS`: Five+ person households

Use Case: Household size stratification for unit type targeting

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Part 3: Age Variables

B01001: Age and Sex

Variable Codes (examples):

- `B01001_003`: Male, 18-19 years
- `B01001_004`: Male, 20 years
- `B01001_005`: Male, 21 years
- `B01001_006`: Male, 22-24 years
- `B01001_007`: Male, 25-29 years
- `B01001_008`: Male, 30-34 years
- `B01001_009`: Male, 35-39 years
- `B01001_010`: Male, 40-44 years
- `B01001_011`: Male, 45-49 years
- `B01001_012`: Male, 50-54 years
- `B01001_013`: Male, 55-59 years
- `B01001_014`: Male, 60-61 years
- `B01001_015`: Male, 62-64 years
- `B01001_016`: Male, 65-66 years
- `B01001_017`: Male, 67-69 years
- `B01001_018`: Male, 70-74 years
- `B01001_019`: Male, 75-79 years
- `B01001_020`: Male, 80-84 years
- `B01001_021`: Male, 85+ years
- (Similar codes for females: `B01001_027` through `B01001_049`)

Description: Age and sex distribution of population
Geography Levels: ZIP, CBSA
Use Case: Age cohort stratification
Derived Metrics:

- `MEDIAN_AGE`: Median age of population
- `PCT_AGE_<BRACKET>`: Percentage in age brackets (e.g., `PCT_AGE_25_44`, `PCT_AGE_65_PLUS`)
- `AGE_STRATIFICATION`: `PRIME_BUYING_AGE`, `ESTABLISHED_BUYERS`, `SENIOR_BUYERS`, `YOUNG_BUYERS`

Age Brackets:

- `<18`: Children
- `18–24`: Young adults
- `25–34`: Prime renting/buying age
- `35–44`: Established renters/buyers
- `45–54`: Established buyers
- `55–64`: Pre-retirement buyers
- `65+`: Senior buyers/renters

Part 4: Education Variables

B15003: Educational Attainment

Variable Codes:

- `B15003_022`: Bachelor's degree
- `B15003_023`: Master's degree
- `B15003_024`: Professional degree
- `B15003_025`: Doctorate degree

Description: Educational attainment for population 25 years and over
Geography Levels: ZIP, CBSA
Use Case: Education cohort stratification
Derived Metrics:

- `PCT_BACHELORS_PLUS`: Percentage with Bachelor's degree or higher
- `PCT_GRADUATE_DEGREE`: Percentage with graduate degree (Master's, PhD, professional)
- `EDUCATION_STRATIFICATION`: `HIGH_EDUCATION`, `MIDDLE_EDUCATION`, `LOW_EDUCATION`

Part 5: Poverty Variables

B17001: Poverty Status

Variable Codes:

- `B17001_002`: Income in the past 12 months below poverty level
- `B17001_031`: Income in the past 12 months at or above poverty level

Description: Poverty status of households
Geography Levels: ZIP, CBSA
Use Case: Poverty stratification for affordable housing targeting
Derived Metrics:

- `PCT_BELOW_POVERTY`: Percentage below federal poverty line
- `POVERTY_STRATIFICATION`: `HIGH_POVERTY`, `MODERATE_POVERTY`, `LOW_POVERTY`

Part 6: Rent Burden Variables

B25070: Gross Rent as Percentage of Income

Variable Codes:

- `B25070_002`: Less than 10.0 percent
- `B25070_003`: 10.0 to 14.9 percent
- `B25070_004`: 15.0 to 19.9 percent
- `B25070_005`: 20.0 to 24.9 percent
- `B25070_006`: 25.0 to 29.9 percent

- `B25070_007`: 30.0 to 34.9 percent
- `B25070_008`: 35.0 to 39.9 percent
- `B25070_009`: 40.0 to 49.9 percent
- `B25070_010`: 50.0 percent or more

Description: Gross rent as percentage of household income
Geography Levels: ZIP, CBSA
Use Case: Rent burden stratification
Derived Metrics:

- `PCT_RENT_BURDENED`: Percentage spending >= 30% of income on rent
- `PCT_SEVERELY_BURDENED`: Percentage spending >= 50% of income on rent
- `RENT_BURDEN_STRATIFICATION`: `LOW_BURDEN`, `MODERATE_BURDEN`, `HIGH_BURDEN`

B25095: Gross Rent as Percentage of Income (Owner-Occupied)

Variable Codes: Similar to B25070 but for owner-occupied units
Description: Gross rent (for owner-occupied units) as percentage of household income
Geography Levels: ZIP, CBSA
Use Case: Owner-occupied affordability analysis

Part 7: Housing Structure Variables

B25024: Units in Structure

Variable Codes:

- `B25024_002`: 1-unit, detached
- `B25024_003`: 1-unit, attached
- `B25024_004`: 2 units
- `B25024_005`: 3-4 units
- `B25024_006`: 5-9 units
- `B25024_007`: 10-19 units
- `B25024_008`: 20-49 units
- `B25024_009`: 50 or more units

Description: Distribution of housing units by structure type
Geography Levels: ZIP, CBSA
Use Case: Supply stratification (SFR vs MF)
Derived Metrics:

- `PCT_1_UNIT`: Percentage of single-unit structures (SFR supply indicator)
- `PCT_2_4_UNITS`: Percentage of 2-4 unit structures (small MF)
- `PCT_5P_UNITS`: Percentage of 5+ unit structures (large MF)

B25003: Tenure

Variable Codes:

- `B25003_002`: Owner-occupied
- `B25003_003`: Renter-occupied

Description: Owner vs renter occupancy
Geography Levels: ZIP, CBSA
Use Case: Demand stratification (owner vs rental demand)
Derived Metrics:

- `PCT_OWNER`: Percentage of owner-occupied units
- `PCT_RENTER`: Percentage of renter-occupied units

Part 8: Composite Tenant Cohorts

Renter Cohorts

PROFESSIONAL_RENTER_COHORT:

- Income: `HIGH_INCOME` (>= 100K median OR >= 40% in 100K+ brackets)
- Rent Burden: `LOW_BURDEN` (<30% of income)
- Education: `HIGH_EDUCATION` (>= 40% Bachelor's+)
- Use Case: Traditional SFR equity offerings

FAMILY_RENTER_COHORT:

- Income: `MIDDLE_INCOME` (50K-100K median OR >= 40% in 50K-150K brackets)
- Rent Burden: `MODERATE_BURDEN` (30-50% of income)
- Household Type: `FAMILY_HOUSEHOLD_COHORT` (>= 60% family households)
- Use Case: Front Yard SFR equity offerings

AFFORDABLE_RENTER_COHORT:

- Income: `LOW_INCOME` (<50K median OR >= 40% in <50K brackets)
- Rent Burden: `HIGH_BURDEN` (>50% of income)
- Education: `LOW_EDUCATION` (<20% Bachelor's+)
- Use Case: Affordable housing offerings

MIXED_RENTER_COHORT:

- Balanced mix of characteristics
- Use Case: General SFR equity offerings

Buyer Cohorts

PROFESSIONAL_BUYER_COHORT:

- Income: `HIGH_INCOME`
- Education: `HIGH_EDUCATION`
- Age: `PRIME_BUYING_AGE` (>= 30% aged 25-44)
- Use Case: Homebuilder finance (horizontal development)

FAMILY_BUYER_COHORT:

- Income: `MIDDLE_INCOME`
- Household Type: `FAMILY_HOUSEHOLD_COHORT`
- Age: `PRIME_BUYING_AGE`
- Use Case: Homebuilder finance (vertical construction)

ESTABLISHED_BUYER_COHORT:

- Income: `HIGH_INCOME`
- Age: `ESTABLISHED_BUYERS` (>= 30% aged 45-64)
- Education: `HIGH_EDUCATION`
- Use Case: Homebuilder finance (land banking)

FIRST_TIME_BUYER_COHORT:

- Income: `MIDDLE_INCOME`
- Age: `YOUNG_BUYERS` (>= 20% aged <25) OR `PRIME_BUYING_AGE`
- Education: `MIDDLE_EDUCATION`
- Use Case: Homebuilder finance (first-time buyer programs)

Part 9: Data Sources and Tables

Source Tables

1. **`SOURCE_PROD.ACS.ACS_LONG_ZCTA5`**:
    - Raw ACS data at ZIP (ZCTA5) level
    - Contains VAR_CODE, VAR_LABEL, VALUE, YEAR
    - Used for: Direct variable extraction
2. **`TRANSFORM_PROD.FACT.FACT_ACS_ZIP_TS`**:
    - Transformed ACS data at ZIP level
    - Contains METRIC_ID, VALUE, DATE_REFERENCE
    - Used for: ZIP-level stratification views
3. **`TRANSFORM_PROD.FACT.FACT_ACS_CBSA_TS`**:
    - Transformed ACS data at CBSA level (aggregated from ZIP)
    - Contains METRIC_ID, VALUE, DATE_REFERENCE
    - Used for: CBSA-level stratification views

Catalog Tables

1. **`ADMIN.CATALOG.DIM_METRIC`**:
    - Central registry of all metrics
    - Contains METRIC_ID, METRIC_NAME, DOMAIN, TAXON, GEOGRAPHY_LEVELS
    - Used for: Metric discovery and validation
2. **`ADMIN.CATALOG.DIM_TAXON`**:

- Taxon classifications (HH_DEMOGRAPHY, HH_AFFORDABILITY, HH_STRUCTURE, HH_LABOR)
- Used for: Metric categorization

Part 10: Metric ID Format

ACS metrics in STRATA follow a standardized METRIC_ID format:

Format: `ACS_<VAR_CODE>`

Examples:

- `ACS_B19013_001`: Median household income
- `ACS_B19001_002`: Households with income <10K
- `ACS_B25070_007`: Rent burden 30.0-34.9 percent

Geography-Specific Metrics:

- ZIP-level: Stored in `FACT_ACS_ZIP_TS` with `ID_ZIP`
- CBSA-level: Stored in `FACT_ACS_CBSA_TS` with `ID_CBSA`

Part 11: Stratification Views

Base Stratification Views

1. **`V_ZIP_STRATIFICATION`**:
   - Base ZIP-level ACS stratification
   - Includes: Income, household type, age, education, poverty, rent burden
   - Location: `ANALYTICS_PROD.MARKETS.V_ZIP_STRATIFICATION`
2. **`V_CBSA_STRATIFICATION`**:
   - Base CBSA-level ACS stratification
   - Mirrors ZIP-level structure
   - Location: `ANALYTICS_PROD.MARKETS.V_CBSA_STRATIFICATION`

Product-Type Stratification Views

1. **`V_PRODUCT_TYPE_ACS_STRATIFICATION_ZIP`**:
   - Product-type-specific stratification (SF, MF, BTR, Affordable)
   - Includes: Supply and demand stratifications for each product type
   - Location: `ANALYTICS_PROD.MARKETS.V_PRODUCT_TYPE_ACS_STRATIFICATION_ZIP`

Tenant Cohort Views

1. **`V_TENANT_COHORT_STRATIFICATION_ZIP`**:
   - Composite tenant cohort view (ZIP level)
   - Includes: Income, household type, age, education, poverty, rent burden cohorts

- Location: `ANALYTICS_PROD.MARKETS.V_TENANT_COHORT_STRATIFICATION_ZIP`

2. **`V_TENANT_COHORT_STRATIFICATION_CBSA`**:
   - Composite tenant cohort view (CBSA level)
   - Mirrors ZIP-level structure
   - Location: `ANALYTICS_PROD.MARKETS.V_TENANT_COHORT_STRATIFICATION_CBSA`

Part 12: Usage Examples

Example 1: Query Income Stratification

```sql
SELECT
    ID_ZIP,
    MEDIAN_HOUSEHOLD_INCOME,
    INCOME_STRATIFICATION,
    PCT_MIDDLE_INCOME,
    PCT_HIGH_INCOME
FROM ANALYTICS_PROD.MARKETS.V_ZIP_STRATIFICATION
WHERE DATE_REFERENCE >= DATEADD(YEAR, -2, CURRENT_DATE())
  QUALIFY ROW_NUMBER() OVER (PARTITION BY ID_ZIP ORDER BY DATE_REFERENCE
DESC) = 1
ORDER BY MEDIAN_HOUSEHOLD_INCOME DESC;
```

Example 2: Query Renter Cohorts

```sql
SELECT
    ID_ZIP,
    ZIP_RENTER_INCOME_COHORT,
    ZIP_RENTER_INCOME_STRATIFICATION,
    ZIP_RENT_BURDEN_STRATIFICATION,
    ZIP_PCT_HIGH_INCOME,
    ZIP_PCT_SEVERELY_BURDENED
FROM ANALYTICS_PROD.MODELED.PROG_SFR_BASE_CURRENT_OUTLOOK_ZIP
WHERE ZIP_RENTER_INCOME_COHORT = 'PROFESSIONAL_RENTER_COHORT'
ORDER BY ZIP_MARKET_RANK_2Y;
```

Example 3: Query Buyer Cohorts

```sql
SELECT
    ID_CBSA,
    MARKET_NAME,
    BUYER_INCOME_COHORT,
    BUYER_HOUSEHOLD_TYPE_COHORT,
    BUYER_AGE_COHORT,
    PCT_HIGH_INCOME,
```

```sql
    PCT_FAMILY_HOUSEHOLDS,
    PCT_AGE_25_44
FROM ANALYTICS_PROD.MODELED.ANCHOR_LOANS_CURRENT_OUTLOOK
WHERE BUYER_INCOME_COHORT = 'PROFESSIONAL_BUYER_COHORT'
ORDER BY MARKET_RANK_2Y;
```

References

- **ACS Data Documentation**: U.S. Census Bureau ACS Documentation
- **STRATA Documentation**:
  - `docs/TENANT_COHORT_STRATEGY.md`: Tenant cohort strategy overview
  - `docs/ACS_VARIABLE_INVENTORY.md`: ACS variable inventory and gap analysis
- **Discovery Results**:
  - `exports/fund_7_q1_2026/metadata/acs_metrics_from_catalog.json`: Metrics from DIM_METRIC catalog
  - `exports/fund_7_q1_2026/metadata/acs_variables_from_source.json`: Variables from source data
  - `exports/fund_7_q1_2026/metadata/acs_coverage_gaps.json`: Gap analysis results

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